

SCHAFFEN AB, Belgium

WMO: 064650

Lat: 50.9969N

Lon: 5.0619E

Elev: 177

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	24.5	13.2	10.8	24.6	17.7	13.5	27.4	25.2	48.7	22.8	46.5	6.2	40	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	86.2	67.3	82.1	65.9	78.8	64.3	69.3	81.7	67.5	78.6	66.0	75.3	6.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	94.5	73.3	63.8	89.1	71.1	62.4	84.7	69.9	33.5	81.8	32.0	78.9	30.8	75.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.3	16.8	15.0	DB	16.6	93.1	5.3	3.7	12.7	95.8	9.6	98.0	6.6	100.1	2.7	102.8
			WB	15.9	72.6	5.1	2.2	12.3	74.2	9.3	75.5	6.5	76.7	2.8	78.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	38.5	39.6	44.6	51.1	57.0	62.3	66.2	64.9	59.7	53.6	45.3	39.8	(6)
	DBStd	11.69	7.56	7.11	6.70	6.93	6.40	5.70	5.91	4.93	5.25	6.19	6.16	7.34	(7)
	HDD50	1442	359	296	190	67	11	0	0	0	0	35	164	320	(8)
	HDD65	4978	821	713	633	418	258	119	55	60	177	354	590	780	(9)
	CDD50	2161	3	3	22	102	229	370	503	463	290	147	24	5	(10)
	CDD65	221	0	0	0	2	11	39	93	58	17	1	0	0	(11)
	CDH74	1998	0	0	0	44	118	365	819	507	137	8	0	0	(12)
	CDH80	558	0	0	0	4	18	91	264	157	24	0	0	0	(13)
(14) Wind	WSAvg	7.3	8.6	8.4	8.1	7.3	7.3	6.7	6.5	6.1	6.3	6.8	7.4	8.1	(14)
(15) Precipitation	PrecAvg	30.9	2.6	2.3	2.2	1.9	2.4	2.9	3.0	3.2	2.3	2.5	2.6	2.9	(15)
	PrecMax	40.6	4.9	5.6	5.3	4.6	5.5	6.6	9.0	8.5	6.9	6.9	4.3	6.1	(16)
	PrecMin	23.6	0.1	0.2	0.1	0.2	0.9	0.6	1.2	0.6	0.6	1.2	0.2	0.7	(17)
	PrecStd	4.6	1.2	1.2	1.1	1.0	1.1	1.3	1.6	2.0	1.5	1.1	1.0	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	59.2	67.2	79.4	83.0	88.8	93.4	90.1	83.7	74.5	63.4	57.0	(19)
		MCWB	51.3	49.0	52.8	59.2	65.1	68.5	69.1	67.9	66.6	62.7	57.1	53.0	(20)
	2%	DB	53.6	54.6	62.5	73.4	77.4	83.1	86.6	84.5	78.0	69.1	59.7	54.5	(21)
		MCWB	50.4	48.4	51.2	56.8	61.1	65.8	67.2	67.2	64.7	60.9	55.0	51.1	(22)
	5%	DB	50.8	51.8	58.0	68.2	74.0	78.7	82.2	79.5	73.4	65.2	56.9	52.2	(23)
		MCWB	47.7	47.6	49.9	54.8	59.7	64.0	66.1	65.6	62.8	58.7	52.9	49.3	(24)
	10%	DB	48.5	49.4	54.7	63.5	69.5	74.4	78.5	75.0	69.1	62.3	54.5	49.8	(25)
		MCWB	45.7	46.0	48.3	52.8	58.2	62.4	64.9	63.5	60.7	57.2	51.3	47.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	51.7	55.1	60.4	67.3	71.6	71.6	72.0	68.5	64.3	58.3	53.9	(27)
		MCDB	54.6	55.8	62.5	74.7	78.8	84.3	85.5	84.3	79.5	71.6	62.0	56.0	(28)
	2%	WB	50.4	49.8	52.8	58.0	63.8	68.0	69.2	68.8	66.1	61.5	55.4	51.6	(29)
		MCDB	53.1	53.1	59.5	69.9	72.8	78.4	82.4	80.5	74.8	67.8	58.8	53.9	(30)
	5%	WB	48.1	48.1	50.9	55.7	61.2	65.6	67.5	66.8	64.0	59.7	53.6	49.5	(31)
		MCDB	50.5	51.0	56.8	66.3	70.1	74.9	79.3	76.4	71.4	64.3	56.5	51.8	(32)
	10%	WB	45.9	46.0	49.1	53.7	59.1	63.5	65.9	65.0	62.0	58.0	51.4	47.6	(33)
		MCDB	48.3	49.3	54.0	62.1	67.7	72.4	75.6	73.0	67.8	61.8	54.3	49.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	10.1	13.5	16.9	16.8	17.4	17.2	16.8	15.8	13.1	9.7	8.2	(35)
		MCDBR	9.1	13.4	20.0	24.5	23.7	24.3	23.8	24.8	22.0	16.8	12.0	8.8	(36)
	5% WB	MCWBR	7.9	9.1	11.3	11.3	10.8	10.3	9.0	10.3	10.3	9.8	8.8	7.4	(37)
		MCWBR	9.2	10.9	16.8	21.9	19.6	20.0	20.9	21.1	19.6	15.2	10.9	8.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.328	0.352	0.385	0.417	0.426	0.423	0.429	0.420	0.391	0.366	0.341	0.326	(40)
		taud	2.385	2.347	2.256	2.194	2.213	2.239	2.227	2.274	2.342	2.400	2.389	2.353	(41)
	Ebn at Noon		218	244	255	260	263	265	261	257	251	235	210	197	(42)
		Edh at Noon	20	27	35	42	43	42	42	38	32	25	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		243	457	825	1315	1571	1706	1610	1373	1027	605	286	189	(44)
	RadStd		33	89	117	147	144	146	193	134	102	49	32	27	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (85 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page